

- The Problem:

I was once asked for my opinion on the following problem:

"I have to sample from a dataset a pool of 120 subjects.

Each subject, enrolled in a given site (S1, S2, S3, S4) has five demographic/clinical characteristics (DC1, DC2, ..., DC5) to be represented in the analysis:

- 1. DC1 - can have six different values (X1 - X6).*
- 2. DC2 - can have two different values (G1, G2)*
- 3. DC3 - can have two different values (E1, E2)*
- 4. DC4 - can have three different values (M1, M2, M3)*
- 5. DC5- can have two different values (C1, C2)*

Subjects have to be randomly selected such that when all cases are pooled:

- i. DC1: A maximum of 70% of cases will come from subjects with DC1=X1*
- ii. DC2: 1/3 of subjects from group G1 and 2/3 from group G2 (i.e., twice as many subjects from group G2 vs. group G1)*
- iii. DC3: At least 20% from the group E1*
- iv. DC4: At least 20% from each group (M1, M2, M3)*
- v. DC5: At least 10% from group C1*
- vi. Site: At least 10% from each site.*

Which procedure or other solution do you know?"

I didn't know any SAS or other statistical procedures, nor was I able to find anything useful across the internet. So I solved the problem on my own with poor theoretical reasoning but rich common sense. The first solution that comes to mind is pretty simple: just randomly sample and check whether all the criteria are met. But this one is computationally heavy, though its theoretical properties are quite nice.

The last solution uses some reasoning from linear programming: to decide whether optimization (in this case, we will be optimizing some kind of a

functional of satisfiability) has really happened, we check how hard it was to move an improvement along one of the dimensions (for example, it can be hard to catch).

- Definitions:

- o Let's define Weak constraint as the requirement of having no lesser than or no more than x subjects with some characteristic. *For example, requirements i, iii, iv, v, vi.*
- o Let's define Strong constraint as the requirement of having exact number of subjects in the sampled dataset. *For example, requirement ii.*
- o $K=120$ – is sample size, original dataset size N is larger, but unknown, it's just some constant.

- Assumptions:

- o Let's denote the set of all the samples, meeting the prespecified requirements, as X . Then the sampling is the process of choosing one of them, i.e. x . We want **to sample them in uniform fashion**, i.e. all the samples have the probability of being chosen, namely $p=1/\text{card}(X)$.
- o I assume that there is at least one sample meeting requirements, i.e. $\text{card}(X) \geq 1$, i.e. it's not necessary to check the **existence**.
- o **Independence of features.** I assume that the features $\{\text{DC1}, \dots, \text{DC5}, \text{Site}\}$ are distributed independently. This a) may be too idealistic, but I know that the investigators usually try to balance subgrouping of subjects; b) facilitates the calculations and helps to make some initial estimates.
- o [I deal with exactly one feature which is strongly constrained, this is DC2].

- **Baseline solution**

- o **Idea (alg1):** 1) shuffle original dataset, 2) choose first K subjects, 3) then check whether criteria are met, 4->1) if not – repeat. Works well, but the probability of meeting a weak constraint (ii) is expected to be extremely low.

Then let's modify this idea (alg2):

- 1) shuffle,
- 2,3) choose first <strong_constraint_number> subjects satisfying 1st strong constraint and then choose subjects satisfying the rest (Namely, 40 and 80),
- 4->1) repeat if not all the criteria are met.
- o **Very simple.**
- o **Absolutely random.**
- o **Works pretty fine if:**
 - Probability of a subject having one of desired group characteristics is equal to one of the weak constraints. For example, if one of weak constraints is (actually it is) “vi. Site: At least 10% from each site”, then the algorithm works well if $p(\text{subject has DC1=X1})=0.1$. We also have other such similar probabilities, predefined by the other weak constraints. Then the probability, that a sample meets all the constraints is $P(\text{sample meets constr (i) \&... \& sample meets constr (vi) | constraint (ii) already met}) = [\text{since features distributed independently}] = P(\text{sample meets constr (i)}) \dots P(\text{sample meets constr (vi)}) = (1 - \text{BinomialCDF}(84 \text{ to } 120, p=1-0.7)) \dots \text{ThreenomialCDF}(x \geq 24, y \geq 24, p1=0.2, p2=0.2) \dots \sim 1$, since all the factors are ~ 1 . This means, that in such cases algorithm will work almost in one repetition.
 - The algorithm also works fine, if probabilities of any feature having one of it's values is equal to $1/(\text{number of feature values})$. Then the factors in the equation above become $\sim 1/2$, and the probability that a sample meets all the criteria is $\sim 1/32$. This means, that it's enough some reasonably small number of times to repeat the algorithm's procedure to obtain the desired sample.
- o **The expected number of required iterations increases significantly, if at least one of probabilities is “imbalanced” for the worse.** For example, if probability $p(\text{subject comes from site S1})=0.07$
- o **Uniformity:** since dataset is shuffled uniformly, i.e. among all $N!$ subjects permutations one is chosen, then the first 40 subjects satisfying first part (ii), and the rest of 80, satisfying the rest of (ii), are chosen to form the desired sample, the obtained sample may be thought of as uniformly chosen, but some additional reasoning is needed to prove or disprove it.
- o **This approach may be easily implemented in few code lines in SAS or Python.**

- “Linear-programming”-like approach (**alg12**)

- o **Idea:**

- 1) Shuffle dataset;
 - 2-3) Choose first <strong_constraint_number> subjects satisfying 1st strong constraint and then choose subjects satisfying the rest (Namely, 40 and 80),
 - 4) create “accumulator” to measure, how “distant” are the current indicators of the current sample from weak constraints satisfaction;
 - 5) iterate by chosen subjects, within each one iterate by unchosen subjects;
 - 6) check whether criteria are met. If so, stop and return formed sample. If no, go to (7);
 - 7) for current pair of (chosen, unchosen) subjects decide, whether the 2nd one can improve current indicators to shorten break between criteria and indicators;
 - If no, then continue iterating by unchosen subjects, i.e. continue (5)
 - If yes
 - then substitute chosen subject by unchosen one,
 - go to the next chosen subject in (5)
 - 8 ->1) go to (1) if didn't succeed.

- *Remark: The decision process in (6) is a Bernoulli trial with probability p_w of success being heuristically defined.*

I defined it as a proportion of weights: some “positive” weights to all of them.

Weights are estimated as abs value of difference between boundary of weak constraint (for example, if constraint prescribes “No more than 70% subjects have $DC1=X1$ ”, then boundary is 0.7) and simple proportion of subjects with characteristic of interest (for example, $X1$) within the whole dataset or chosen subjects.

Then p_w may be estimated as proportion of

Numerator=weights of those characteristics, that 1) were improved & has lower proportion than constraint's boundary & needed to improve; 2) were improved & have greater proportion, than constraint's boundary & needed to improve; 3) half (!) of weights of characteristics that worsen & have greater proportion than contrs. boundary & currently in surplus.

and the Denominator= weights of all the other characteristics, that are weakly constrained.

At the same time, whatever the proportion is, the decision is negative if we cannot improve the current deficit of characteristics.

- o **Quicker than baseline solution** (if probabilities of having characteristics are small)
 - o **“Partly random”**
 - o **Complicated**
 - o **Almost uniform.** It has some uniform fashion, since the original dataset is shuffled, then “the first possible” (if possible) sample, that satisfies criteria, is chosen, but this also means that the algorithm has no temptation to exceed criteria. But this may be partly fixed by

additional heuristics when dealing with weights. *[Comparison using JS and KL divergence of marginal distributions is provided in the notebook]*

- o **May do poor work if the number of samples, meeting all the criteria, is noticeably small –** but I didn't investigate this.
- **To check these approaches, I generated a dataset of 240/ 480 subjects with some prespecified (but not significantly low) probabilities of having characteristics (19 in total, 12 of them are of interest)** .
- At the beginning I also tried to find some simple SAS or general solutions, but I didn't succeed 😊